Activity Pulse

An AI Assistant for Project Monitoring and Document Workflows

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1.125 Agentic Engineering October 29, 2026

Project information is fragmented

The borrower implements. The Bank provides implementation support and oversight, including document review and no-objection where required.

For Bank task teams: TTLs, procurement and operations staff

Information domain	Source	What the team learns
Formal records	STEP	Procurement stage, cost, method, review type
Formal records	Aide Memoire / ISR	Agreed actions, owners, dates / progress, ratings, risks
Correspondence	Email	Requests, responses and open questions
Supporting documents	Activity files / SharePoint	Documents under review and their versions

What awaits Bank review, and what awaits borrower action?

Each clearance document adds manual work: download, date-name, upload to SharePoint, review, then tell the team.

Actionable status for each activity

SYNTHETIC EXAMPLE / As of October 28, 2026

National Data Center / Oversight consultancy

CS-xx QCBS Prior review

STEP: Shortlisting Last email: Oct 20 (8 days ago)

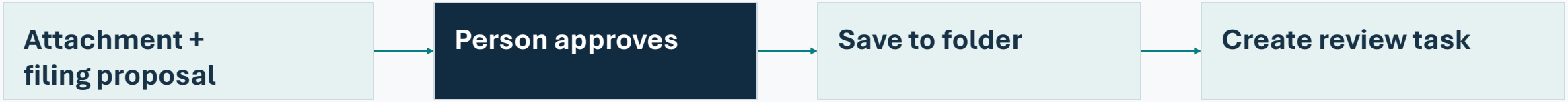
Pending: Bank comments on Shortlisting Evaluation Report v2

Type: Request Proposed owner: Bank procurement specialist Confidence: medium

Evidence: Email Oct 20, STEP row Documents: latest SER v2, activity folder

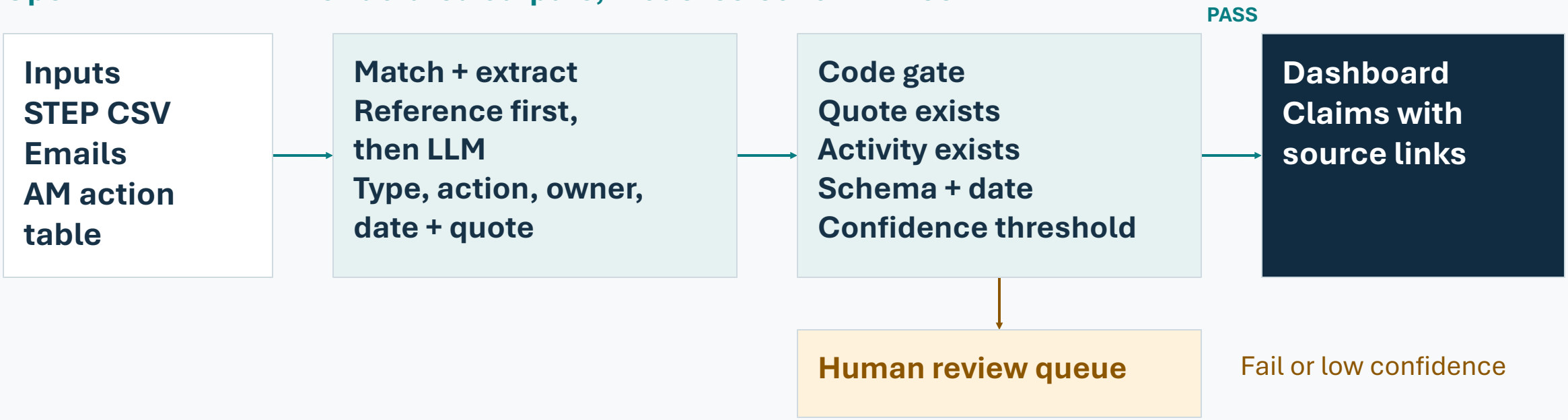
Pending review of a clearance request

This request establishes neither a completed action nor a discrepancy with STEP.



A controlled AI workflow with human oversight

Python (workflow) / SQLite (data) / Streamlit (interface) / pytest (tests)
OpenAI API: LLM with structured outputs; model selection in Week 1



A quote match checks traceability. It does not verify the meaning of the claim.



A realistic, bounded prototype

CORE

Match emails to activities and extract pending actions with evidence. Show an activity dashboard. Complete one approved local filing workflow.

Designed to complement STEP and collaboration tools with proposed actions and source evidence.

Priority	Planned capability
Supporting	AM action table, overdue commitments, latest version per document stream
Stretch	ISR extraction, discrepancy detection, version comparison, Russian/Tajik
Future	Live STEP, Outlook and SharePoint, multiple projects, notifications

Synthetic data only. No confidential Bank or government material.

No outgoing emails or STEP updates. Human-approved saves only, with no overwrites and an audit log.

Build, test, demonstrate

Four-week implementation plan

Week	Milestone
1	Dataset + answer key Command-line matching / extraction
2	Validation gate + dashboard Midpoint matching evaluation
3	Document intake + human review queue
4	Evaluation, README + demo Supporting features if time allows

Success targets

≥90% correct activity matching
≥80% extraction accuracy per field
100% source traceability
0 unapproved saves or overwrites

Measure errors and review referrals separately.
These are prototype targets, not guarantees.
Gather feedback from 2–3 WB colleagues.

Demo

Load samples, find an open clearance request, approve filing,
create a review task, then open the evidence from the dashboard.

A1 Data model and document handling

Component / Procurement activity (STEP reference) / Document stream / Versions

Separate streams preserve simultaneous documents within one activity.
The dashboard opens the activity folder or the latest version of each stream.

Record	Key content
Email and extraction	Value, source quote, statement type, confidence and source reference
AM action and task	Commitment, proposed owner, target date and review status
Audit log	Approval, append-only save and resulting review task

File naming and local folder

YYYY-MM-DD_ActivityRef_DocType_vN_fromGov.ext
/Component/ActivityRef/DocType/

A2 Interpretation and evaluation

Every extracted statement has a type

Request **Commitment** **Prediction** **Completed action**

Only a completed action with supporting evidence can prompt a discrepancy review against STEP. A request or prediction is not evidence of completion.

Measure	How to report it
Matching: target ≥90%	Correct matches / all emails. Also report precision and coverage. Report errors and review referrals. Never silently drop an email.
Extraction: target ≥80% per field	Score each field, including statement type, against human labels. Report counts, denominators and missing or abstained fields.
Traceability: target 100%	Test source links and quote presence for every displayed claim. Evaluate semantic correctness separately against human labels.
Filing: zero violations	Test approval enforcement, no overwrites and complete save logs.

A3 Resources, risks and design logic

Minimal stack

Python, SQLite, Streamlit
OpenAI API with structured output
pytest for gates and filing behavior

Resources

Laptop, API access and a capped test budget
Synthetic fixtures and a hand-labeled key
Feedback from 2–3 WB colleagues

Main risks and mitigations

Email interpretation: build and test first.
Unclear owner: route to human review.
Python/API learning: keep the stack small.

Synthetic dataset

2 components, 5 activities, 15–20 emails
3–5 document versions, 1 AM action table

Axiomatic Design: need to design

Need	Design choice
Know status	Extraction + activity card
Trust it	Validation + evaluation
Know next action	Proposed owner + review
Safe filing	Human approval gate

Core needs no live institutional access. Live Graph / SharePoint integration is future work.

A4 Sources, disclosure and presentation plan

Authorship and AI assistance

Project concept and workflow: Nekruz Khamidov. Outline refinement: ChatGPT / MIT AI (Parley). Slide drafting and layout: OpenAI Codex.
All example records are synthetic. This proposal reports no implemented results.

References and planned tools

PS06 requirements: github.com/onexi/ps06
Python: docs.python.org/3/ SQLite: sqlite.org/docs.html
Streamlit: docs.streamlit.io pytest: docs.pytest.org
OpenAI structured outputs: platform.openai.com/docs/guides/structured-outputs
Future integration: learn.microsoft.com/graph/api/resources/sharepoint
World Bank: Project Cycle; STEP overview; Procurement Regulations, Annex II

Course concepts: J. R. Williams, A. Sanchez, "Agentic System Architectures," 1.125 lecture, Oct 27, 2026; "Building Blocks of Design" and "AI Accelerated Design RunBook" (Axiomatic Design), same lecture.

Speaking plan: title 0:10, then 0:45 / 1:00 / 1:10 / 0:40 / 0:50 = 4:35.

Leave 0:25 for transitions. Appendix supports the PDF and questions only.